Notes on Weak Identification

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1 Basics

We introduce the estimation methods of two stage least squares (2SLS), limited-information maximum likelihood (LIML), full-information maximum likelihood (FIML), and the k-class estimator. Those methods are explained in the simultaneous equation models. This section is mainly from Chow (1983) and the original ones are due to Theil's book and papers. In the following, we adopt the conventional notations where $P_A = A(A'A)^{-1}A'$, $M_A = I - P_A$ for a generic matrix A.

We consider a s with the structural form (SF), and there are $p+1$ endogenous variables and $r+1$ exogenous variables. Its specification is as follows.

$$By_t + \Gamma_1 y_{t-1} + \cdots + \Gamma_p y_{t-p} + \Gamma_0 z_t + \Gamma_{p+1} z_{t-1} + \cdots + \Gamma_{p+r} z_{t-r} = \varepsilon_t, \quad t = 1, \cdots, n \tag{1}$$

where there are G structural equations in the t-th observation, meaning that y_t, ε_t are G-dimensional vectors of exogenous variables, $\mathbb{E}(\varepsilon_t) = 0$, $\mathbb{E}(\varepsilon_t \varepsilon_t') = \Sigma$, and $\mathbb{E}(\varepsilon_t \varepsilon_{t-k}') = 0$ for integer $k \neq 0$. Then the reduced form (RF) of (1) is

$$y_t = \Pi_1 y_{t-1} + \cdots + \Pi_p y_{t-p} + \Pi_0 z_t + \Pi_{p+1} z_{t-1} + \cdots + \Pi_{p+r} z_{t-r} + v_t \tag{2}$$

Let $x_t = (y_{t-1}', \cdots, y_{t-p}', z_t', z_{t-1}', \cdots, z_{t-r}')'$ and $\Gamma = [\Gamma_1, \cdots, \Gamma_{p+r}]$ and $\Pi = [\Pi_1, \cdots, \Pi_{p+r}]$, Γ is a $G \times K$ matrix where $K = G(p+r+1)$, so (1) and (2) can be rewritten as

$$By_t + \Gamma x_t = \varepsilon_t \tag{3}$$

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$$y_t = -B^{-1}\Gamma x_t + B^{-1}\varepsilon_t = \Pi x_t + v_t \tag{4}$$

let $Y' = [y_1, \cdots, y_n]$, $X' = [x_1, \cdots, x_n]$, and E', V' are similarly defined, so the SF and RF

$$BY' + \Gamma X' = E' \tag{5}$$

$$Y' = \Pi X' + B^{-1}E' = \Pi X' + V' \tag{6}$$

note that the i-th row of $[B, \Gamma]$ denotes the coefficients of the i-th structural equation. Transpose (5) gives

$$YB' + X\Gamma' = E \tag{7}$$

Let $\bar{\beta}_i$ and $\bar{\gamma}_i$ denote the i-th column of B and Γ, then the i-the equation is

$$Y\bar{\beta}_i + X\bar{\gamma}_i = \varepsilon_i \tag{8}$$

where ε_i is the error term in the i-th equation for all observations, to be distinguished with ε_t which represents the error term in the t-th observation for all equations. ε_i is $n \times 1$ and ε_t is $G \times 1$.

Consider a limited-information method of estimation of (8), we assume a prior restriction to ensure identification, many elements of $\bar{\beta}_i$ and $\bar{\gamma}_i$ are zero, so corresponding columns of Y and X can be omitted from (8). If one further assumes $\beta_{ii} = -1$, then

$$-y_i + Y_i\beta_i + X_i\gamma_i = \varepsilon_i \tag{9}$$

Let $W_i = [Y_i, X_i]$ and $\delta_i = (\beta_i', \gamma_i')'$, then

$$y_i = W_i\delta_i + \varepsilon_i \tag{10}$$

Here we have the endogenous problem which means W_i and ε_i are correlated for Y_i being correlated with ε_i. Fortunately, X_i are actually instrument variables for Y_i, so the 2SLS estimator is formed through

$$\widehat{Y}_i = X_i(X_i'X_i)^{-1}X_i'Y_i = P_{X_i}Y_i, \quad \widehat{W}_i = [\widehat{Y}_i, X_i] = P_{X_i}W_i \tag{11}$$

$$\widehat{\delta}_{\text{2SLS}} = (\widehat{W}_i'\widehat{W}_i)^{-1}\widehat{W}_i'y_i \tag{12}$$

1.1 Limited-Information Maximum Likelihood (LIML)

We consider the 1st structural equation in (9), i.e., $i = 1$,

$$-y_1 + Y_1\beta_1 + X_1\gamma_1 == \varepsilon_1$$

1.2 The k-Class Estimator

In this part we omit the label "1" for simplicity. So the model is

$$y = Y\beta + X\gamma + \varepsilon$$

The k-class estimator suggested by Theil (1958) includes 2SLS and LIML as special cases. The estimator obtain estimates for β and γ by solving the following equation,

$$\begin{bmatrix} Y'Y - k\widehat{V}'\widehat{V} & Y'X \\ X'Y & X'X \end{bmatrix} \begin{bmatrix} \widehat{\beta}(k) \\ \widehat{\gamma}(k) \end{bmatrix} = \begin{bmatrix} Y' - k\widehat{V}' \\ X' \end{bmatrix} y \tag{13}$$

where $\widehat{V} = M_X Y$ and here X are actually the instrument for Y. In more complicated structural equation system, $\widehat{V}$ should be considered as the (valid) first stage regression residual satisfying the exogenous condition.

Remark: If $k = 0$, then $\widehat{\beta}(k)$ is actually the OLS estimator whereas it's clearly inconsistent due to the endogeneity (the choice of k to ensure consistency is illustrated in Theorem 1); if $k = 1$, then the k-class estimator goes back to 2SLS; if $k = \mu$, then it goes back to LIML.

Theorem 1. If $k \xrightarrow{\mathbb{P}} 1$, then the k-class estimators are consistent for δ and have the same limiting distribution.

Proof of Theorem 1. By (13) we have

$$\begin{aligned} &(Y'Y - k\widehat{V}'\widehat{V})\widehat{\beta} + Y'X\widehat{\gamma} = Y'y - K\widehat{V}'y \\ &X'Y\widehat{\beta} + X'X\widehat{\gamma} = X'y \end{aligned}$$

□

2 Weak Instrument and Its Asymptotics

Some notations. $P_W = W(W'W)^{-1}W'$ and $M_W = I - P_W$ and $W^{\perp} = M_X W$ for a generic matrix W. One will immediately obtain that P_W, M_W are symmetric and idemponent. And

$$W'P_W = W'W(W'W)^{-1}W' = W'$$
$$P_W W = W(W'W)^{-1}W'W = W$$

The P_W is a projection matrix that project to W.

2.1 Rothenberg's Decomposition

Let y, Y be $T \times 1$ and Z be $T \times k$ (k instruments), and (y, Y, T) are observed

$$y = Y\beta + u, \quad u_i \overset{iid}{\sim} \mathcal{N}(0, \sigma_{uu})$$
$$Y = Z\Pi + V, \quad V_i \overset{iid}{\sim} \mathcal{N}(0, \sigma_{VV}), \quad u \perp V$$

Consider 2SLS. For the first step, $\widehat{Y} = Z(Z'Z)^{-1}Z'Y = P_Z Y$,

$$\widehat{\beta}^{\text{2SLS}} = (\widehat{Y}'\widehat{Y})^{-1}\widehat{Y}'y = (Y'P_Z'P_Z Y)^{-1}Y'P_Z'y = \frac{Y'P_Z y}{Y'P_Z Y}$$

Define the concentration parameter to be

$$\mu^2 = \frac{\Pi' Z' Z \Pi}{\sigma_{VV}} \tag{14}$$

Then

$$\widehat{\beta}^{\text{2SLS}} = \frac{Y'P_Z(Y\beta + u)}{Y'P_Z Y} = \beta + \frac{Y'P_Z u}{Y'P_Z Y}$$

So

$$\begin{aligned}
\mu(\widehat{\beta}^{\text{2SLS}} - \beta) &= \mu \cdot \frac{(Z\Pi + V)'P_Z u}{(Z\Pi + V)'P_Z Y} = \mu \cdot \frac{(Z\Pi + V)'P_Z u}{(Z\Pi + V)'P_Z Y} = \mu \cdot \frac{\Pi' Z' P_Z u + V' P_Z u}{\Pi' Z' P_Z Y + V' P_Z Y} \\
&= \mu \cdot \frac{\Pi' Z' u + V' P_Z u}{\Pi' Z' Y + V' P_Z Y} = \mu \cdot \frac{\Pi' Z' u + V' P_Z u}{\Pi' Z'(Z\Pi + V) + V' P_Z (Z\Pi + V)} \\
&= \mu \cdot \frac{\Pi' Z' u + V' P_Z u}{\Pi' Z' Z\Pi + \Pi' Z' V + V' Z \Pi + V' P_Z V} = \frac{(\Pi' Z' Z \Pi)^{1/2}}{\sigma_{VV}^{1/2}} \frac{\Pi' Z' u + V' P_Z u}{\Pi' Z' Z \Pi + 2\Pi' Z' V + V' P_Z V} \\
&= \frac{\sigma_{uu}^{1/2}}{\sigma_{VV}^{1/2}} \cdot \frac{\frac{\Pi' Z' u}{(\Pi' Z' Z \Pi)^{1/2}\sigma_{uu}^{1/2}} + \frac{V' P_Z u}{(\Pi' Z' Z \Pi)^{1/2}\sigma_{uu}^{1/2}}}{1 + \frac{2\Pi' Z' V}{\Pi' Z' Z \Pi} + \frac{V' P_Z V}{\Pi' Z' Z \Pi}}
\end{aligned}$$

$$= \frac{\sigma_{uu}^{1/2}}{\sigma_{VV}^{1/2}} \cdot \frac{\frac{\Pi' Z' u}{(\Pi' Z' Z \Pi)^{1/2} \sigma_{uu}^{1/2}} + \frac{V' P_Z u}{\mu \sigma_{VV}^{1/2} \sigma_{uu}^{1/2}}}{1 + \frac{2 \Pi' Z' V}{\mu (\Pi' Z' Z \Pi)^{1/2} \sigma_{VV}^{1/2}} + \frac{V' P_Z V}{\mu^2 \sigma_{VV}}}$$

$$= \frac{\sigma_{uu}^{1/2}}{\sigma_{VV}^{1/2}} \cdot \frac{\zeta_u + \frac{S_{Vu}}{\mu}}{1 + \frac{2\zeta_V}{\mu} + \frac{S_{VV}}{\mu^2}} = (\frac{\sigma_{uu}}{\sigma_{VV}})^{1/2} (\frac{\zeta_u + S_{Vu}/\mu}{1 + 2(\zeta_V/u) + (S_{VV}/u^2)})$$

where $\zeta_u = \frac{\Pi' Z' u}{(\Pi' Z' Z \Pi)^{1/2} \sigma_{uu}^{1/2}} \sim \mathcal{N}(0,1)$, $\zeta_V = \frac{\Pi' Z' V}{(\Pi' Z' Z \Pi)^{1/2} \sigma_{VV}^{1/2}} \sim \mathcal{N}(0,1)$, $S_{Vu} = V' P_Z u/(\sigma_{VV} \sigma_{uu})^{1/2}$, $S_{VV} = V' P_Z V/\sigma_{VV}$ and concentration parameter μ is defined as

$$\mu^2 = \frac{\Pi' Z' Z \Pi}{\sigma_{VV}}$$

<u>Note. 1</u> $\mathrm{Cov}(\zeta_u, \zeta_V) = \rho$.

2.2 Pre-Test for Weak Instrument

In this section we review the very popular Stock-Yogo test for weak instrument which refers to Stock and Yogo (2005). Let's consider

$$y = Y\beta + X\gamma + u \tag{15}$$

$$Y = Z\Pi + X\Phi + v \tag{16}$$

Y is $T \times n$ matrix of n endogenous variables and T denotes the number of observations, and X contains K_1 exogenous variables, Z is the IV matrix which contains K_2 instrument variables ($K_2 \geq n$). Let $\underline{X} = (Y, X)$ denote the regressors in the first stage, $\underline{Y} = [y, Y]$ to compress all the endogenous variables and $\underline{Z} = [Z, X]$ to include all the exogenous variables. Note $M^{\perp}$ have a special meaning that is the residual of regressing on the exogenous regressors X.

Let X_t denote the t-th observation of X which is stored in a column vector form, i.e., $X_t = (X_{t1}, X_{t2}, \cdots, X_{tK_1})'$ and Z_t, V_t are similarly defined, and $\underline{Z}_t = (Z_t', X_t')'$. And define

$$\mathbb{E}[\begin{pmatrix} u_t \\ V_t \end{pmatrix} (u_t, V_t')] = \begin{bmatrix} \sigma_{uu} & \Sigma_{uV} \\ \Sigma_{uV} & \Sigma_{VV} \end{bmatrix} = \Sigma \tag{17}$$

$$\mathbb{E}(\underline{Z}_t \underline{Z}_t') = \begin{bmatrix} Q_{ZZ} & Q_{ZX} \\ Q_{XZ} & Q_{XX} \end{bmatrix} = Q \tag{18}$$

The k-class estimator $\widehat{\beta}(k)$ is obtained by solving

$$\begin{bmatrix} Y'Y - k\widehat{V}'\widehat{V} & Y'X \\ X'Y & X'X \end{bmatrix} \begin{bmatrix} \widehat{\beta}(k) \\ \widehat{\gamma}(k) \end{bmatrix} = \begin{bmatrix} Y' - k\widehat{V}' \\ X' \end{bmatrix} y \tag{19}$$

where $\widehat{V} = (I_T - \underline{Z}(\underline{Z}'\underline{Z})^{-1}\underline{Z}')Y = M_{\underline{Z}}Y$. Moreover, we can write the k-class estimator in a more compact form

$$(\widehat{\beta}'(k), \widehat{\gamma}'(k))' = [\underline{X}'(I - kM_{\underline{Z}})\underline{X}]^{-1}[\underline{X}'(I - kM_{\underline{Z}})y] \tag{20}$$

Proof of (20). Firstly it can be proved that

$$M_{\underline{Z}} = M_{Z^\perp} M_X \tag{21}$$

so $X'M_{\underline{Z}}Y = (M_{\underline{Z}}X)'Y = (M_{Z^\perp}M_X X)'Y = 0$ for $M_X X = 0$. Similarly, $Y'M_{\underline{Z}}X = 0$ and $X'M_{\underline{Z}}X = 0$. $\square$

Then one is able to derive the estimator for βm

$$\widehat{\beta}(k) = [Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp]^{-1}[Y^{\perp\prime}(I - kM_{Z^\perp})y^\perp] \tag{22}$$

Proof of (22). Recall that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} (A - BD^{-1}C)^{-1} & -(A - BD^{-1}C)^{-1}BD^{-1} \\ -D^{-1}C(A - BD^{-1}C)^{-1} & D^{-1} + D^{-1}C(A - BD^{-1}C)^{-1}BD^{-1} \end{bmatrix}$$

for generic matrices A, B, C, D. Thus by the above formula,

$$\begin{bmatrix} Y'Y - k\widehat{V}'\widehat{V} & Y'X \\ X'Y & X'X \end{bmatrix}^{-1} = \begin{bmatrix} (Y^{\perp\prime}Y^\perp - k\widehat{V}'\widehat{V})^{-1} & -(Y^{\perp\prime}Y^\perp - k\widehat{V}'\widehat{V})^{-1}Y'X(X'X)^{-1} \\ .. & .. \end{bmatrix}$$

Thus by (19), we have

$$\widehat{\beta}(k) = (Y^{\perp\prime}Y^\perp - k\widehat{V}'\widehat{V})^{-1}(Y^{\perp\prime}y^\perp - k\widehat{V}'y)$$

If one shows that $\widehat{V} = M_{\underline{Z}}Y = M_{Z^\perp}Y^\perp = M_{Z^\perp}M_X Y$, then the proof is done. And indeed by (21), $M_{\underline{Z}} = M_{Z^\perp}M_X$, so we derive (22). $\square$

If we consider the Wald test for $H_0 : \beta = \beta_0$, then the Wald test statistic is

$$W(k) = \frac{[\widehat{\beta}(k) - \beta_0]'[Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp][\widehat{\beta}(k) - \beta_0]}{n\widehat{\sigma}_{uu}(k)} \tag{23}$$

where $\widehat{\sigma}_{uu}(k) = \widehat{u}^\perp(k)'\widehat{u}^\perp(k)/(T - K_1 - n)$ and $\widehat{u}^\perp(k) = y^\perp - Y^\perp\widehat{\beta}(k)$.

Proof of (23). Let's first calculate the variance of $\widehat{\beta}(k)$

$$\begin{aligned}\widehat{\beta}(k) - \beta_0 &= [Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp]^{-1}[Y^{\perp\prime}(I - kM_{Z^\perp})M_X(Y\beta_0 + X\gamma + u)] - \beta_0 \\ &= [Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp]^{-1}[Y^{\perp\prime}(I - kM_{Z^\perp})M_X(X\gamma + u)]\end{aligned}$$

let $S = Y^{\perp\prime}(I - kM_{Z^\perp})M_X$, note SY is symmetric, thus

$$\begin{aligned}&\mathrm{Var}(\widehat{\beta}(k) - \beta_0) = \mathbb{E}[(SY)^{-1}(Su)(u'S')(SY)^{-1}] \\ &= \sigma_{uu}[Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp]^{-1}Y^{\perp\prime}(I - kM_{Z^\perp})M_X(I - kM_{Z^\perp})Y^\perp[Y^{\perp\prime}(I - kM_{Z^\perp})Y^\perp]^{-1}\end{aligned}$$

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3 Anderson Rubin test

4 Wald, LR, LM tests in GMM Methods

Recall

$$\begin{aligned}y &= Y\beta + X\gamma + \varepsilon \\ Y &= Z\Pi + X\Gamma + V\end{aligned}$$

5 Inference for structural parameters in weak IV

5.1 LM test, K-tatistic, Kleibergen (2002)

Consider the limited information simultaneous equation model

$$\begin{aligned}y &= Y\beta + X\gamma + \varepsilon \\ Y &= Z\Pi + X\Gamma + V\end{aligned}$$

where y is $T \times 1$ and Y is $T \times m$ where y and Y are endogenous variable, and Z is $T \times k_Z$ of exogenous variables. ε is $T \times 1$ of structural errors and V is $T \times m$ of reduced form errors.

Assumption 5.1. We made the following convergence assumptions.

(i) $(1/T)(\varepsilon V)'(\varepsilon V) \xrightarrow{\mathbb{P}} \Sigma$ where

$$\Sigma = \begin{pmatrix} \sigma_{\varepsilon\varepsilon} & \sigma_{\varepsilon V} \\ \sigma_{V\varepsilon} & \Sigma_{VV} \end{pmatrix}$$

(ii) $1/T(ZX)^\intercal(ZX) \xrightarrow{\mathbb{P}} Q$, with Q a positive definite $k \times k$ matrix.

$$Q = \begin{pmatrix} Q_{ZZ} & Q_{ZX} \\ Q_{XZ} & Q_{XX} \end{pmatrix}$$

(iii)

$$\frac{1}{\sqrt{T}}(ZX)^\intercal(\varepsilon V) \xrightarrow{d} \begin{pmatrix} \psi_{Z\varepsilon} & \psi_{ZV} \\ \psi_{X\varepsilon} & \psi_{XV} \end{pmatrix}$$

and

$$\text{vec} \begin{pmatrix} \psi_{Z\varepsilon} & \psi_{ZV} \\ \psi_{X\varepsilon} & \end{pmatrix}$$

For the testing problem

$$H_0 : \beta = \beta_0 \qquad H_a : \beta \neq \beta_0$$

The Anderson-Rubin (AR) statistic is

$$\text{AR}(\beta_0) = \frac{\frac{1}{k}(y - Y\beta_0)^\intercal P_X (y - Y\beta_0)}{\frac{1}{T-k}(y - Y\beta_0)^\intercal M_X (y - Y\beta_0)} \tag{24}$$

and we have

$$\text{AR}(\beta_0) \xrightarrow{d,H_0} \frac{\chi^2(k)}{k} \tag{25}$$

One deficiency of the AR test is the degree of freedom of the limiting distribution is equal to the number of instruments k, which may lead to the low power of the test. And Kleibergen (2002) proposed the following

$$K(\beta_0) = \frac{(y - Y\beta_0)^\intercal P_{\tilde{Y}(\beta_0)}(y - Y\beta_0)}{\frac{1}{T-k}(y - Y\beta_0)^\intercal M_X(y - Y\beta_0)} \tag{26}$$

where

$$\begin{aligned}
\tilde{Y}(\beta_0) &= X\tilde{\Pi}(\beta_0) \\
\tilde{\Pi}(\beta_0) &= (X^\intercal X)^{-1}X^\intercal \left[Y - (y - Y\beta_0)\frac{S_{\varepsilon V}(\beta_0)}{S_{\varepsilon\varepsilon}(\beta_0)}\right] \\
S_{\varepsilon\varepsilon}(\beta_0) &= \frac{1}{T-k}(y - Y\beta_0)^\intercal M_X(y - Y\beta_0) \\
S_{\varepsilon V}(\beta_0) &= \frac{1}{T-k}(y - Y\beta_0)^\intercal M_X Y
\end{aligned}$$

5.2 Conditional likelihood ratio (CLR) test, Moreira (2003).

Let $P_A = A(A'A)^{-1}A'$ and $M_A = I - P_A$, both are idempotent.

Structural first stage and reduced form second stage are

$$\underbrace{y_1}_{n\times 1} = \underbrace{Y_2}_{n\times l}\underbrace{\beta}_{l\times 1} + \underbrace{X_1}_{n\times r}\underbrace{\gamma_1}_{r\times 1} + u, \qquad \mathbb{E}(u) = 0, \quad \mathrm{Cov}(Y_2, u) \neq 0 \tag{27}$$

$$\underbrace{Y_2}_{n\times l} = \underbrace{X_2}_{n\times k}\underbrace{\Pi}_{k\times l} + \underbrace{X_1}_{n\times r}\underbrace{\overline{\Gamma}}_{r\times l} + V_2, \qquad k \geq l, \quad [X_1, X_2] \text{ full rank } (r+k) \tag{28}$$

Let $Z = M_{X_1}X_2$, the "real instrument", and $\Gamma = \overline{\Gamma} + (X_1'X_1)^{-1}X_1'X_2\Pi$, so

$$\begin{aligned}
Y_2 &= (M_{X_1} + P_{X_1})X_2\Pi + X_1(\Gamma - (X_1'X_1)^{-1}X_1'X_2\Pi) + V_2 \\
&= Z\Pi + X_1\Gamma + V_2 \\
y_1 &= Z\Pi\beta + X_1\Gamma\beta + V_2\beta + X_1\gamma_1 + u \\
&= Z\Pi\beta + X_1(\Gamma\beta + \gamma_1) + (V_2\beta + u) \\
&= Z\Pi\beta + X_1\delta + v_1
\end{aligned}$$

To summarize, the reduced form equations are

$$\begin{cases} y_1 = Z\Pi\beta + X_1\delta + v_1 \\ Y_2 = Z\Pi + X_1\Gamma + V_2 \end{cases} \tag{29}$$

Let's denote $V = [v_1, V_2]$, so V is $n \times (l+1)$ and each row of V are normally i.i.d. with mean 0, and covariance matrix $\Omega \in \mathbb{R}^{(l+1)\times(l+1)}$. We write

$$Y = [y_1, Y_2] = Z\Pi \underbrace{[\beta, I_l]}_{A'} + X_1 \underbrace{[\delta, \Gamma]}_{\Phi} + \underbrace{[v_1, V_2]}_{V}$$

For convenience we define $\Pi_* = \Pi A'$, so

$$Y = Z\Pi_* + X_1\Phi + V$$

Note $Z'YD$, where $D \in \mathbb{R}^{(l+1)\times(l+1)}$ is full-rank, is also an invariant sufficient statistic. We choose $D = D_0$,

$$D_0 = [b_0, \Omega^{-1}A_0], \quad b_0 = [1, -\beta_0']', \quad A_0 = [\beta_0, I_l]'$$

and define

$$\begin{aligned} S &= Z'Yb_0 \\ T &= Z'Y\Omega^{-1}A_0 \end{aligned}$$

Let i-the row of V be denoted by $V_{i\cdot}$. Due to $V_{i\cdot}$ are i.i.d. $\mathcal{N}(0, \Omega)$, the joint density of $V_{i\cdot}$ is

$$f(V) = (2\pi)^{-n/2}|\Omega|^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^{n} V_{i\cdot}'\Omega^{-1}V_{i\cdot}\right)$$

Note $\sum_{i=1}^{n} V_{i\cdot}'\Omega^{-1}V_{i\cdot} = \mathrm{tr}(V\Omega^{-1}V')$, so the log-likelihood is

$$\begin{aligned} L(\Pi_*, \Phi, \Omega) &= -\frac{n}{2}\log|\Omega| - \frac{1}{2}\mathrm{tr}(\Omega^{-1}V'V) \\ &= -\frac{n}{2}\log|\Omega| - \frac{1}{2}\mathrm{tr}[\Omega^{-1}(Y - Z\Pi_* - X_1\Phi)'(Y - Z\Pi_* - X_1\Phi)] \end{aligned}$$

Letting $\partial\mathcal{L}/\partial\Phi = 0$ gives estimate $\hat{\Phi}$, i.e.,

$$\mathrm{tr}[\Omega^{-1}(Y - Z\Pi_* - X_1\hat{\Phi})'X_1] = 0 \implies \hat{\Phi} = (X_1'X_1)^{-1}X_1'(Y - Z\Pi_*)$$

Plugging $\hat{\Phi}$ to the log-likelihood will concentrate out Φ, note

$$\hat{V} = Y - Z\Pi_* - X_1\hat{\Phi} = M_{X_1}(Y - Z\Pi_*) = M_{X_1}V$$

where the last step is due to $M_{X_1}X_1 = 0$. So the concentrated log-likelihood can be written as

$$L(Y;\beta,\Pi,\Omega) = -\frac{n}{2}\log|\Omega| - \frac{1}{2}\operatorname{tr}(\Omega^{-1}V'M_{X_1}V) \tag{30}$$

$$= -\frac{n}{2}\log|\Omega| - \frac{1}{2}\operatorname{tr}(\Omega^{-1}(Y - Z\Pi_*)'M_{X_1}(Y - Z\Pi_*)) \tag{31}$$

where $A = [\beta, I_l]'$ and $\Pi_* = \Pi A'$. We want to concentrate out Π. Maximizing the above is equivalent to

$$\min_{\Pi_*:\Pi_* b=0} \frac{1}{2}\operatorname{tr}(\Omega^{-1}(Y - Z\Pi_*)'M_{X_1}(Y - Z\Pi_*))$$

The Lagrangian is, with the multiplier Λ,

$$\mathcal{L} = \frac{1}{2}\operatorname{tr}(\Omega^{-1}(Y - Z\Pi_*)'M_{X_1}(Y - Z\Pi_*)) + \operatorname{tr}(\Lambda b'\Pi_*')$$

The FOC for Π_* is, (note $\frac{\partial}{\partial x}(f(x)'Af(x)) = 2\frac{\partial}{\partial x}f(x)'Af(x))$

$$-Z'M_{X_1}(Y - Z\hat{\Pi}_*)\Omega^{-1} + \Lambda b' = 0$$

Note $M_{X_1}Z = Z$, we have

$$Z'Z\hat{\Pi}_* = Z'Y - \Lambda b'\Omega$$

$$\hat{\Pi}_* = (Z'Z)^{-1}Z'Y - (Z'Z)^{-1}\Lambda b'\Omega$$

And $\hat{\Pi}_* b = 0$ gives

$$\Lambda = \frac{Z'Yb}{b'\Omega b}$$

So we have

$$\hat{\Pi}_* = (Z'Z)^{-1}Z'Y - (Z'Z)^{-1}Z'Yb(b'\Omega b)^{-1}b'\Omega$$

Then we need to plug $\hat{\Pi}_*$ back into (31). Note if we set $X = [X_1, X_2]$

$$Z = M_{X_1}X_2 \implies M_{X_1} = M_X + N_Z$$

and $M_X Z = (M_{X_1} - N_Z)Z = Z - Z = 0$, so

$$\Omega^{-1}(Y - Z\hat{\Pi}_*)'M_{X_1}(Y - Z\hat{\Pi}_*) = \Omega^{-1}(Y - Z\hat{\Pi}_*)'(M_X + N_Z)(Y - Z\hat{\Pi}_*)$$

$$= \Omega^{-1} Y' M_X Y + \Omega^{-1} (Y - Z\hat{\Pi}_*)' N_Z (Y - Z\hat{\Pi}_*)$$

and note $Z'Y - Z'Z\hat{\Pi}_* = \Lambda b'\Omega$, we have

$$\begin{aligned}
(Y - Z\hat{\Pi}_*)' N_Z (Y - Z\hat{\Pi}_*) &= (Y - Z\hat{\Pi}_*)' Z (Z'Z)^{-1} Z' (Y - Z\hat{\Pi}_*) \\
&= \Omega b \Lambda' (Z'Z)^{-1} \Lambda b' \Omega = \frac{\Omega b b' Y' Z (Z'Z)^{-1} Z' Y b b' \Omega}{(b'\Omega b)^2} \\
&= \frac{\Omega b b' Y' N_Z Y b b' \Omega}{(b'\Omega b)^2}
\end{aligned}$$

So (31) is further concentrated as

$$\begin{aligned}
L_c(Y; \beta, \Omega) &= -\frac{n}{2} \log |\Omega| - \frac{1}{2} \operatorname{tr}(\Omega^{-1} Y' M_X Y + \Omega^{-1} (Y - Z\hat{\Pi}_*)' N_Z (Y - Z\hat{\Pi}_*)) \\
&= -\frac{n}{2} \log |\Omega| - \frac{1}{2} \left[\operatorname{tr}(\Omega^{-1} Y' M_X Y) + \operatorname{tr}\left(\frac{b b' Y' N_Z Y b b' \Omega}{(b'\Omega b)^2} \right) \right] \\
&= -\frac{n}{2} \log |\Omega| - \frac{1}{2} \left[\operatorname{tr}(\Omega^{-1} Y' M_X Y) + \frac{b' Y' N_Z Y b}{b'\Omega b} \right] \qquad (32)
\end{aligned}$$

Maximizing $L_c(Y; \beta, \Omega)$ is equivalent to

$$\min_b \frac{b' Y' N_Z Y b}{b'\Omega b} = \frac{(\Omega^{1/2} b)' \Omega^{-1/2} Y' N_Z Y \Omega^{-1/2} (\Omega^{1/2} b)}{(\Omega^{1/2} b)' (\Omega^{1/2} b)} = \bar{\lambda}_{\min}$$

where $\bar{\lambda}_{\min}$ is the minimum eigenvalue of $\Omega^{-1/2} Y' N_Z Y \Omega^{-1/2}$, which gives us the maximum of L_c,

$$L_c(Y; \hat{\beta}, \Omega) = -\frac{n}{2} \log |\Omega| - \frac{1}{2} \left[\operatorname{tr}(\Omega^{-1} Y' M_X Y) + \bar{\lambda}_{\min} \right]$$

And (32) implies the likelihood ratio

$$LR_0 = \frac{b_0 Y' N_Z Y b_0}{b_0' \Omega b_0} - \bar{\lambda}_{\min}$$

Let

$$\begin{aligned}
&\bar{S} = (Z'Z)^{-1/2} S (b_0' \Omega b_0)^{-1/2}, \quad S = Z' Y b_0, \quad \bar{S} \in \mathbb{R}^{k \times 1} \\
&\bar{T} = (Z'Z)^{-1/2} T (A_0' \Omega^{-1} A_0)^{-1/2}, \quad T = Z' Y \Omega^{-1} A_0, \quad \bar{T} \in \mathbb{R}^{k \times l}
\end{aligned}$$

then

$$\bar{S}' \bar{S} = \frac{S' (Z'Z)^{-1} S}{b_0 \Omega b_0} = \frac{b_0' Y' Z (Z'Z)^{-1} Z' Y b_0}{b_0 \Omega b_0}$$

so

$$LR_0 = \bar{S}'\bar{S} - \bar{\lambda}_{\min}$$

By the way, $\bar{\lambda}_{\min}$ is also the smallest eigenvalue of

$$(\bar{S}, \bar{T})'(\bar{S}, \bar{T}) = \begin{pmatrix} \bar{S}'\bar{S} & \bar{S}'\bar{T} \\ \bar{T}'\bar{S}' & \bar{T}'\bar{T} \end{pmatrix}$$

To see this, note

$$(\bar{S}, \bar{T}) = (Z'Z)^{-1/2}Z'Y\Omega^{-1/2}\underbrace{[\Omega^{1/2}b_0(b_0'\Omega b_0)^{-1/2}, \Omega^{-1/2}A_0(A_0'\Omega^{-1}A_0)^{-1}]}_{J}$$

The matrix $J \in \mathbb{R}^{(l+1)\times(l+1)}$ is special, it's an orthogonal matrix. Let's check

$$J'J = \begin{pmatrix} I_1 & 0_{1\times l} \\ 0_{l\times 1} & I_l \end{pmatrix} = I_{l+1}$$

for $[\Omega^{1/2}b_0(b_0'\Omega b_0)^{-1/2}]'[\Omega^{-1/2}A_0(A_0'\Omega^{-1}A_0)^{-1}] = [b_0'A_0(A_0'\Omega^{-1}A_0)^{-1}]/(b_0'\Omega b_0)^{1/2} = 0_{1\times l}$ due to $b_0'A_0 = 0_{1\times l}$. Then we have

$$\begin{aligned}(\bar{S}, \bar{T})'(\bar{S}, \bar{T}) &= [(Z'Z)^{-1/2}Z'Y\Omega^{-1/2}J]'[(Z'Z)^{-1/2}Z'Y\Omega^{-1/2}J] \\ &= J'\Omega^{-1/2}Y'Z(Z'Z)^{-1}Z'Y\Omega^{-1/2}J \\ &= J'(\Omega^{-1/2}Y'N_ZY\Omega^{-1/2})J\end{aligned}$$

So $(\bar{S}, \bar{T})'(\bar{S}, \bar{T})$ and $\Omega^{-1/2}Y'N_ZY\Omega^{-1/2}$ share the same set of eigenvalues.

- Simplifications

  (i) The β is scalar, i.e., $l = 1$. Then $(\bar{S}, \bar{T})'(\bar{S}, \bar{T})$ is 2×2 matrix. By quadratic equation algebra. The solution has a closed form.

  (ii) When $l = 1$ and in the just-identified case, so that $l = k = 1$, the LRT becomes AR test

5.3 AMS, Andrews, Moreira, and Stock (2006)

Suppose

$$\underbrace{y_1}_{n\times 1} = \underbrace{y_2}_{n\times 1}\beta + X\gamma_1 + u \tag{33}$$

$$\underbrace{y_2}_{n\times 1} = \underbrace{\tilde{Z}}_{n\times k}\pi + \underbrace{X}_{n\times p}\xi_1 + v_2 \tag{34}$$

where observed variables are $(y_1, y_2, X, \tilde{Z})$, unobserved errors are (u, v_2); unknown parameters are $(\beta, \gamma_1, \pi, \xi_1)$. The equation (34) implies

$$\begin{aligned}
y_2 &= (I - P_X)\tilde{Z}\pi + P_X\tilde{Z}\pi + X\xi_1 + v_2 \\
&= Z\pi + X\left[(X'X)^{-1}X'\tilde{Z}\pi + \xi_1\right] + v_2 \\
&= Z\pi + X\xi + v_2 \\
y_1 &= (Z\pi + X\xi + v_2)\beta + X\gamma_1 + u \\
&= Z\pi\beta + X(\beta\xi + \gamma_1) + \beta v_2 + u \\
&= Z\pi\beta + X\gamma + v_1
\end{aligned}$$

In matrix form

$$Y = Z\pi a' + X\eta + V$$

5.4 Weak GMM, Kleibergen (2005).

Parameter of interest $\theta = (\theta_1, \cdots, \theta_m)' \in \mathbb{R}^m$ and $f : \mathbb{R}^m \to \mathbb{R}^{k_f}$. We have moment condition

$$\mathbb{E}(f(\theta_0, Y_t)) = 0, \quad t = 1, \cdots, T, \quad k_f \geq m$$

To estimate θ, we use Hansen's GMM.

$$Q(\theta) = f_T(\theta, Y)'\hat{V}_{ff}(\theta)^{-1}f_T(\theta, Y)$$

5.5 HAC-IV, Moreira and Moreira (2019).

Some Kronecker product algebra, let's say $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{m\times m}$

$$(A \otimes B)' = A' \otimes B' \quad (A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$$

$$
\begin{aligned}
&(A \otimes B)(C \otimes D) = AC \otimes BD \\
&A \otimes B \otimes C = A \otimes (B \otimes C) \\
&\operatorname{tr}(A \otimes B) = \operatorname{tr}(A) \operatorname{tr}(B) \\
&|A \otimes B| = |A|^m |B|^n \\
&\text{eigen-property}
\end{aligned}
$$

The model is simpler than Moreira (2003). One endogenous variable ($l = 1$), no other control variables ($r = 0$) which can be justified by "projection arguments" in Andrews, Moreira, and Stock (2006).

$$
\begin{aligned}
\underbrace{y_1}_{n\times 1} &= \underbrace{y_2}_{n\times 1} \beta + u \\
\underbrace{y_2}_{n\times 1} &= \underbrace{Z}_{n\times k} \underbrace{\pi}_{k\times 1} + v_2
\end{aligned}
$$

Note

$$
y_1 = (Z\pi + v_2)\beta + u = Z\pi\beta + \underbrace{(v_2\beta + u)}_{v_1}
$$

then the reduced-form model is

$$
Y = [y_1, y_2] = Z\pi \underbrace{[\beta, 1]}_{a'} + \underbrace{[v_1, v_2]}_{V}
$$

Let

$$
\begin{aligned}
&P_1 = Z(Z'Z)^{-1/2}, \quad P_1 \in \mathbb{R}^{n\times k} \\
&[P_1, P_2] \in \mathcal{O}_n, \quad \mathcal{O}_n \text{ is the group of } n \times n \text{ orthogonal matrices}
\end{aligned}
$$

Consider tests based on $R = P_1'Y$, specifically,

$$
\begin{aligned}
&R = (Z'Z)^{-1/2} Z'Z\pi a' + (Z'Z)^{-1/2} Z'V = \mu a' + (Z'Z)^{-1/2} Z'V, \quad R \in \mathbb{R}^{k\times 2} \\
&\mu = (Z'Z)^{1/2}\pi, \quad \mu \in \mathbb{R}^{k\times 1}
\end{aligned}
$$

where $(Z'Z)^{-1/2}Z'V$ is assumed to be normally distributed with zero mean and a known variance Σ. To proceed, let

$$
\bar{R} = vec(R) = vec(\mu a' + (Z'Z)^{-1/2} Z'V)
$$

$$\mathbb{E}(\bar{R}) = vec(\mu a') = (a \otimes I_k)vec(\mu) = (a \otimes I_k)\mu$$
$$\mathrm{Var}(\bar{R}) = \Sigma \in \mathbb{R}^{2k\times 2k} \quad \text{by assumption}$$

due to $vec(ABC) = (C' \otimes A)vec(B)$. Let

$$b_0 = (1, -\beta_0)'$$
$$C_{\beta_0} = [(b_0' \otimes I_k)\Sigma(b_0 \otimes I_k)]^{-1/2}$$
$$M_{\beta_0} = [(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1/2}$$

Then we consider a 1-1 transformation of $\bar{R}$ by pre-multiplying $\bar{R}$ with a full rank matrix W as follows,

$$\begin{pmatrix} S \\ T \end{pmatrix} = W\bar{R} = \begin{pmatrix} A \\ B \end{pmatrix} \bar{R} = \begin{pmatrix} C_{\beta_0}(b_0' \otimes I_k) \\ M_{\beta_0}(a_0' \otimes I_k)\Sigma^{-1} \end{pmatrix} \bar{R}$$

for (S, T) has the variance

$$\begin{pmatrix} A \\ B \end{pmatrix} \Sigma(A'B') = \begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix} \tag{35}$$

and we calculate

$$A\Sigma A' = C_{\beta_0}(b_0' \otimes I_k)\Sigma(b_0 \otimes I_k)C_{\beta_0} = I_k, \quad B\Sigma B' = I_k$$
$$A\Sigma B' = C_{\beta_0}(b_0' \otimes I_k)\Sigma\Sigma^{-1}(a_0 \otimes I_k)M_{\beta_0} = C_{\beta_0}(b_0'a_0 \otimes I_k)M_{\beta_0} = 0$$

So S and T are independent. And

$$\mathbb{E}(S) = C_{\beta_0}(b_0' \otimes I_k)(a \otimes I_k)\mu = (\beta - \beta_0)C_{\beta_0}\mu, \quad \text{for } b_0'a = (1, -\beta_0)(\beta, 1)' = \beta - \beta_0$$
$$\mathbb{E}(T) = M_{\beta_0}(a_0' \otimes I_k)\Sigma^{-1}(a \otimes I_k)\mu = \underbrace{[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1/2}(a_0' \otimes I_k)\Sigma^{-1}(a \otimes I_k)}_{D_\beta}\mu = D_\beta\mu$$

so the joint density of $(S', T')'$ is

$$f_{\beta,\mu}(s,t) = (2\pi)^{-k/2}\exp\left(-\frac{\|s - (\beta-\beta_0)C_{\beta_0}\mu\|^2}{2}\right)(2\pi)^{-k/2}\exp\left(-\frac{\|t - D_\beta\mu\|^2}{2}\right)$$
$$= f^S_{\beta,\mu}(s) \times f^T_{\beta,\mu}(t)$$

(a) The Anderson-Rubin test.

$$AR = S'S \sim \chi^2(k)$$

(b) Likelihood ratio test.

Note $\bar{R} \sim \mathcal{N}((a \otimes I_k)\mu, \Sigma)$, so

$$L_n(\beta, \mu) = -\frac{1}{2}(\bar{R} - (a \otimes I_k)\mu)'\Sigma^{-1}(\bar{R} - (a \otimes I_k)\mu)$$

Concentrating out μ implies

$$\frac{\partial}{\partial \mu} L_n = (a' \otimes I_k)\Sigma^{-1}(\bar{R} - (a \otimes I_k)\mu) = 0$$

So

$$\hat{\mu}(\beta) = [(a' \otimes I_k)\Sigma^{-1}(a \otimes I_k)]^{-1}(a' \otimes I_k)\Sigma^{-1}\bar{R}$$

Substituting back to L_n gives

$$\begin{aligned}
L_n^c(\beta) &= -\frac{1}{2}(\bar{R} - (a \otimes I_k)\hat{\mu})'\Sigma^{-1}(\bar{R} - (a \otimes I_k)\hat{\mu}) \\
&= -\frac{1}{2}\bar{R}'\Sigma^{-1}\bar{R} + [(a \otimes I_k)\hat{\mu}]'\Sigma^{-1}\bar{R} - \frac{1}{2}[(a \otimes I_k)\hat{\mu}]'\Sigma^{-1}[(a \otimes I_k)\hat{\mu}] \\
&= -\frac{1}{2}\bar{R}'\Sigma^{-1}\bar{R} + \hat{\mu}'(a' \otimes I_k)\Sigma^{-1}\bar{R} - \frac{1}{2}\hat{\mu}(a' \otimes I_k)\Sigma^{-1}\bar{R} \\
&= -\frac{1}{2}\bar{R}'\Sigma^{-1}\bar{R} + \frac{1}{2}\hat{\mu}'(a' \otimes I_k)\Sigma^{-1}\bar{R} \\
&= -\frac{1}{2}\bar{R}'\Sigma^{-1}\bar{R} + \frac{1}{2}\bar{R}'\Sigma^{-1}(a \otimes I_k)[(a' \otimes I_k)\Sigma^{-1}(a \otimes I_k)]^{-1}(a' \otimes I_k)\Sigma^{-1}\bar{R} \\
&= -\frac{1}{2}\bar{R}'\Sigma^{-1}\bar{R} + \frac{1}{2}(\Sigma^{-1/2}\bar{R})'N_{\Sigma^{-1/2}(a \otimes I_k)}\Sigma^{-1/2}\bar{R}
\end{aligned}$$

and note $T = [(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1/2}(a_0' \otimes I_k)\Sigma^{-1}\bar{R}$

$$\begin{aligned}
T'T &= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1}(a_0' \otimes I_k)\Sigma^{-1}\bar{R} \\
&= \bar{R}'\Sigma^{-1/2}N_{\Sigma^{-1/2}(a \otimes I_k)}\Sigma^{-1/2}\bar{R}
\end{aligned}$$

so the likelihood ratio test is

$$\begin{aligned}
LR &= 2[\max_{\beta} L_n^c(\beta) - L_n^c(\beta_0)] \\
&= \max_{a}(\Sigma^{-1/2}\bar{R})'N_{\Sigma^{-1/2}(a \otimes I_k)}\Sigma^{-1/2}\bar{R} - T'T
\end{aligned}$$

(c) LM test.

We need to calculate $\frac{\partial}{\partial\beta}L_n^c(\beta)$. Let $X = \Sigma^{-1/2}(a\otimes I_k) \in \mathbb{R}^{2k\times k}$, $P = (X'X)^{-1} \in \mathbb{R}^{k\times k}$, we have $dP = -P(dX'\cdot X + X'dX)P$ due to the following reason,

$$\begin{aligned}
d((X'X)P) = dI = 0 \implies d(X'X)\cdot P + (X'X)dP = 0 \\
dP = -(X'X)^{-1}d(X'X)\cdot P \\
= -P(dX'\cdot X + X'dX)P
\end{aligned}$$

Note

$$\begin{aligned}
dN_X &= d(XPX') = dX\cdot PX' - X[P(dX'\cdot X + X'dX)P]X' + XPdX' \\
\dot{X} &= \frac{\partial X}{\partial\beta} = \frac{\partial\Sigma^{-1/2}(a\otimes I_k)}{\partial\beta} = \Sigma^{-1/2}(e_1\otimes I_k) \in \mathbb{R}^{2k\times k} \\
\frac{\partial N_X}{\partial\beta} &= \dot{X}PX' - X[P(\dot{X}'X + X'\dot{X})P]X' + XP\dot{X}' \\
&= \dot{X}PX' - XP\dot{X}'XPX' - XPX'\dot{X}PX' + XP\dot{X}' \\
&= (I_{2k} - XPX')\dot{X}PX' + XP\dot{X}'(I_{2k} - XPX') \\
&= A + A'
\end{aligned}$$

where $A \in \mathbb{R}^{2k\times 2k}$. Therefore

$$\begin{aligned}
\frac{\partial L_n^c(\beta)}{\partial\beta} &= \frac{1}{2}\bar{R}'\Sigma^{-1/2}(A + A')\Sigma^{-1/2}\bar{R} \\
&= \frac{1}{2}\bar{R}'\Sigma^{-1/2}A\Sigma^{-1/2}\bar{R} + \frac{1}{2}\bar{R}'\Sigma^{-1/2}A'\Sigma^{-1/2}\bar{R}, \quad \text{“scalar”} \\
&= \bar{R}'\Sigma^{-1/2}A'\Sigma^{-1/2}\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a\otimes I_k)[(a'\otimes I_k)\Sigma^{-1}(a\otimes I_k)]^{-1}(e_1'\otimes I_k)\Sigma^{-1/2}M_{\Sigma^{-1/2}(a\otimes I_k)}\Sigma^{-1/2}\bar{R}
\end{aligned}$$

and clearly,

$$\begin{aligned}
&\frac{\partial L_n^c(\beta_0)}{\partial\beta} = \bar{R}'\Sigma^{-1}(a_0\otimes I_k)[(a_0'\otimes I_k)\Sigma^{-1}(a_0\otimes I_k)]^{-1}(e_1'\otimes I_k)\Sigma^{-1/2}M_{\Sigma^{-1/2}(a_0\otimes I_k)}\Sigma^{-1/2}\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a_0\otimes I_k)[(a_0'\otimes I_k)\Sigma^{-1}(a_0\otimes I_k)]^{-1}(e_1'\otimes I_k)\Sigma^{-1/2}M_{\Sigma^{-1/2}(a_0\otimes I_k)}\Sigma^{-1/2}((e_1, a_0 - \beta_0 e_1)\otimes I_k)\bar{R}
\end{aligned}$$

for $(e_1, a_0 - \beta_0 e_1) = I_2$ implies $(e_1, a_0 - \beta_0 e_1)\otimes I_k = I_{2k}$. Define

$$[\underbrace{X_1}_{2k\times k}, \underbrace{X_2}_{2k\times k}] = [\Sigma^{-1/2}(e_1\otimes I_k), \Sigma^{-1/2}(a_0\otimes I_k)] = \Sigma^{-1/2}[e_1, a_0]\otimes I_k$$

so the derivative of the log-likelihood under the null is

$$\begin{aligned}
\frac{\partial L_n^c(\beta_0)}{\partial \beta} &= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} X_1' M_{X_2}[X_1, X_2 - \beta_0 X_1]\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} X_1' M_{X_2}[X_1, -\beta_0 X_1]\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} X_1' M_{X_2} X_1 [I_k, -\beta_0 I_k]\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} X_1' M_{X_2} X_1 (b_0' \otimes I_k)\bar{R}
\end{aligned}$$

and note

$$\begin{aligned}
([X_1, X_2]'[X_1, X_2])^{-1} &= \left[([e_1, a_0]' \otimes I_k)\Sigma^{-1}[e_1, a_0] \otimes I_k\right]^{-1} \\
&= ([e_1, a_0]^{-1} \otimes I_k)\Sigma([e_1, a_0]'^{-1} \otimes I_k) \\
&= \left(\begin{bmatrix} 1 & -\beta_0 \\ 0 & 1 \end{bmatrix} \otimes I_k\right) \Sigma \left(\begin{bmatrix} 1 & 0 \\ -\beta_0 & 1 \end{bmatrix} \otimes I_k\right) \\
&= \begin{pmatrix} b_0' \otimes I_k \\ e_2' \otimes I_k \end{pmatrix} \Sigma \left(b_0 \otimes I_k, e_2 \otimes I_k\right)
\end{aligned}$$

On the other hand, by the 2×2 block matrix inverse

$$\begin{aligned}
([X_1, X_2]'[X_1, X_2])^{-1} &= \begin{pmatrix} X_1'X_2 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{pmatrix}^{-1} \\
= \begin{pmatrix} [X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1]^{-1} & \cdot \\ \cdot & \cdot \end{pmatrix} &= \begin{pmatrix} (X_1'M_{X_2}X_1)^{-1} & \cdot \\ \cdot & \cdot \end{pmatrix}
\end{aligned}$$

So we have

$$C_{\beta_0}^{-2} = (b_0' \otimes I_k)\Sigma(b_0 \otimes I_k) = (X_1'M_{X_2}X_1)^{-1} \implies C_{\beta_0}^2 = X_1'M_{X_2}X_1$$

so

$$\begin{aligned}
\frac{\partial L_n^c(\beta_0)}{\partial \beta} &= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} C_{\beta_0}^2 (b_0' \otimes I_k)\bar{R} \\
&= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k)[(a_0' \otimes I_k)\Sigma^{-1}(a_0 \otimes I_k)]^{-1} C_{\beta_0} S \\
&= \bar{R}'\Sigma^{-1}(a_0 \otimes I_k) D_{\beta_0}^{-2} C_{\beta_0} S \\
&= T' D_{\beta_0}^{-1} C_{\beta_0} S
\end{aligned}$$

So conditional on T,

$$\frac{\partial L_n^c(\beta_0)}{\partial \beta} = (C_{\beta_0} D_{\beta_0}^{-1} T)' S \sim \mathcal{N}(0, T' D_{\beta_0}^{-1} C_{\beta_0}^2 D_{\beta_0}^{-1} T)$$

and the LM test should be constructed as

$$\begin{aligned} LM &= S'(C_{\beta_0} D_{\beta_0}^{-1} T)[(C_{\beta_0} D_{\beta_0}^{-1} T)'(C_{\beta_0} D_{\beta_0}^{-1} T)]^{-1}(C_{\beta_0} D_{\beta_0}^{-1} T)' S \\ &= S' N_{C_{\beta_0} D_{\beta_0}^{-1} T} S \end{aligned}$$

(d) MM1 test.

In terms of $\bar{R}$.

Consider

$$h_\Lambda(s,t) = \int f_{\beta,\mu}(s,t) d\Lambda(\beta,\mu)$$

and the conditional log-likelihood ratio is

$$\frac{h_\Lambda(s,t)}{f_{\beta_0}^S(s)}$$

Note $(S', T')' = W\bar{R}$, so

$$f_{\beta,\mu}(s,t) = f_{\beta,\mu}^{\bar{R}}(\bar{r}) \cdot |W^{-1}|$$

for $|W\Sigma W| = I$ implying $|W^{-1}| = \Sigma^{1/2}$. So we can just consider $f_{\beta,\mu}^{\bar{R}}(\bar{r})$ and $|W^{-1}|$ as constant will be absorbed into critical value,

$$\bar{R} \sim \mathcal{N}((a \otimes I_k)\mu, \Sigma)$$

Set a prior for (β, μ),

- $\beta \sim \mathcal{N}(\beta_0, \sigma_b^2)$.
- $\mu \sim \mathcal{N}(0, \sigma^2 \Phi)$, $\mu \in \mathbb{R}^k$.

so

$$(a \otimes I_k)\mu \sim \mathcal{N}(0, \sigma^2 (a \otimes I_k) \Phi (a' \otimes I_k))$$

Due to the independence of $\bar{R}$'s mean and its error variance,

$$\bar{R} \mid \beta \sim \mathcal{N}(0, \Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k))$$

so the density of $\bar{R} \mid \beta$ is given by

$$(2\pi)^{-k}|\Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k)|^{-1/2} \exp\left\{-\frac{1}{2}\bar{r}'[\Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k)]^{-1}\bar{r}\right\}$$

so integrating the above over β's distribution gives

$$\begin{aligned} h_1(r;\sigma^2) = (2\pi)^{-k-1/2} \int_{-\infty}^{\infty} & \left|\Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k)\right|^{-1/2} \\ & \times \exp\left\{-\frac{\bar{r}'[\Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k)]^{-1}\bar{r}}{2} - \frac{(\beta-\beta_0)^2}{2\sigma_b^2}\right\} d\beta \end{aligned}$$

There is some subtleties on the denominator $f^S f^T$ and critical value and the idea of conditional inference, conditional on the sufficient statistic.

<u>In terms of (S,T)</u>.

Note

$$\begin{aligned} \begin{pmatrix} S \\ T \end{pmatrix} &= W\bar{R} = \begin{pmatrix} C_{\beta_0}(b_0' \otimes I_k) \\ M_{\beta_0}(a_0' \otimes I_k)\Sigma^{-1} \end{pmatrix} \bar{R} \\ \mathrm{Var}(W\bar{R}) &= W(\Sigma + \sigma^2(a \otimes I_k)\Phi(a' \otimes I_k))W' \\ &= W\Sigma W' + \sigma^2[W(a \otimes I_k)]\Phi[W(a \otimes I_k)]' \\ &= I_{2k} + \sigma^2 \begin{pmatrix} C_{\beta_0}(b_0' \otimes I_k)(a \otimes I_k) \\ M_{\beta_0}(a_0' \otimes I_k)\Sigma^{-1}(a \otimes I_k) \end{pmatrix} \Phi \begin{pmatrix} C_{\beta_0}(b_0' \otimes I_k)(a \otimes I_k) \\ M_{\beta_0}(a_0' \otimes I_k)\Sigma^{-1}(a \otimes I_k) \end{pmatrix}' \\ &= I_{2k} + \sigma^2 \begin{pmatrix} (\beta-\beta_0)C_{\beta_0} \\ D_\beta \end{pmatrix} \Phi \begin{pmatrix} (\beta-\beta_0)C_{\beta_0} \\ D_\beta \end{pmatrix}' \\ &= I_{2k} + \sigma^2 \begin{pmatrix} (\beta-\beta_0)^2 C_{\beta_0}\Phi C_{\beta_0}' & (\beta-\beta_0)C_{\beta_0}\Phi D_\beta' \\ (\beta-\beta_0)D_\beta\Phi C_{\beta_0}' & D_\beta\Phi D_\beta' \end{pmatrix} \\ &\equiv \Psi_{\beta,\sigma} \end{aligned}$$

(e) MM2 test.

Define $l_\beta = (c_\beta, d_\beta)'$, where

$$c_\beta = (\beta - \beta_0)(b_0'\Omega b_0)^{-1/2}$$
$$d_\beta = a'\Omega^{-1}a_0(a_0'\Omega^{-1}a_0)^{-1/2}$$

and adopt reparameterization

$$\tan\theta = \frac{d_\beta}{c_\beta}, \quad \theta \in [-\pi, \pi]$$

The above reparametrization implies an implicit function $\beta = \beta(\theta)$. We set

- $\theta \sim \text{Unif}[-\pi, \pi]$.
- $\mu \mid \theta \sim \mathcal{N}(0, \|l_{\beta(\theta)}\|^{-2}\zeta\Phi)$

Like MM1,

$$\bar{R} \mid \theta \sim \mathcal{N}(0, \Sigma + \zeta\|l_{\beta(\theta)}\|^{-2}(a \otimes I_k)\Phi(a' \otimes I_k))$$

with density conditional on θ,

$$(2\pi)^{-k}\left|\Sigma + \zeta\|l_{\beta(\theta)}\|^{-2}(a \otimes I_k)\Phi(a' \otimes I_k)\right|^{-1/2}\exp\left\{-\frac{\bar{r}'(\Sigma + \zeta\|l_{\beta(\theta)}\|^{-2}(a \otimes I_k)\Phi(a' \otimes I_k))^{-1}\bar{r}}{2}\right\}$$

where $a = a_{\beta(\theta)} = (\beta(\theta), 1)$, implying

$$\begin{aligned} h_2(r; \zeta) = (2\pi)^{-k-1}\int_{-\pi}^{\pi} &\left|\Sigma + \zeta\|l_{\beta(\theta)}\|^{-2}(a_{\beta(\theta)} \otimes I_k)\Phi(a'_{\beta(\theta)} \otimes I_k)\right|^{-1/2} \\ &\exp\left\{-\frac{\bar{r}'(\Sigma + \zeta\|l_{\beta(\theta)}\|^{-2}(a_{\beta(\theta)} \otimes I_k)\Phi(a'_{\beta(\theta)} \otimes I_k))^{-1}\bar{r}}{2}\right\} d\theta \end{aligned}$$

5.6 Conditional linear combination (CLC) test

The general question.

- A sequence of model index by sample size T, the data follows distribution $F_T(\theta, \gamma)$.
- Parameter of interest $\theta \in \Theta \subseteq \mathbb{R}^p$, nuisance $\gamma \in \Gamma \subseteq \mathbb{R}^l$, and γ is consistently estimable.
- Testing problem $H_0 : \theta = \theta_0$.
- Assume we observe three objects:

– A k-dimensional vector $g_T(\theta_0)$: "moments".

– A $k \times p$ matrix $\Delta g_T(\theta_0)$: some transformation of the Jacobian of g_T w.r.t. θ.

– An estimate $\hat{\gamma}$ for γ.

- Assume $\forall\,(\theta,\gamma) \in \Theta \times \Gamma$, we have

$$\hat{\gamma} \xrightarrow{\mathbb{P}} \gamma \quad \text{along } F_T(\theta,\gamma)$$

and

$$\begin{pmatrix} g_T(\theta_0) \\ \Delta g_T(\theta_0) \end{pmatrix} \xrightarrow{d} \begin{pmatrix} g \\ \Delta g \end{pmatrix}$$

where

$$\begin{pmatrix} g \\ \text{vec}(\Delta g) \end{pmatrix} \sim N\left(\begin{pmatrix} m \\ \text{vec}(\mu) \end{pmatrix}, \begin{pmatrix} I & \Sigma_{g\theta} \\ \Sigma_{\theta g} & \Sigma_{\theta\theta} \end{pmatrix}\right) \tag{36}$$

in which $\mu \in \mathbb{M}$, $m = m(\theta,\theta_0,\gamma) \in \mathcal{M}(\mu,\gamma) \subseteq \mathbb{R}^k$, that is, the set of possible value of m is determined by μ and γ.[1] Further assume $\Sigma_{\theta g}$ and $\Sigma_{\theta\theta}$ are continuous function of γ, and are thus consistently estimable, and $\Sigma_{\theta g} \in \mathbb{R}^{kp\times k}$.

- The null $\theta = \theta_0$ implies $m = 0$. So testing problem becomes

$$H_0 : m = 0, \mu \in \mathbb{M} \qquad \text{against} \qquad H_1 : m \in \mathcal{M}(\mu) \setminus \{0\}, \mu \in \mathbb{M}$$

Examples.

(i) Weak IV.

Linear instrumental variable with a single endogenous regressor, k , in reduced form:

$$Y = Z\pi\beta + V_1,$$
$$X = Z\pi + V_2,$$

[1] To gain some intuition, the m will be different under the null and alternative, therefore its range depends on some "description" of the identification strength, μ, and of course depends on the nuisance, γ. In weak jump setting, μ denotes the size of the jump, and γ can be the pre/post-announcement volatilities.

for Z is $T \times k$, all of Y, X, V_1 and V_2 are $T \times 1$. The identifying assumption is

$$\mathbb{E}[V_{1,t} Z_t] = \mathbb{E}[V_{2,t} Z_t] = 0$$

So moment condition is

$$f_t(\beta) = (Y_t - X_t \beta) Z_t$$

where Z_t is the transpose of row t of Z, and we have the identifying assumption

$$\mathbb{E}_\beta[f_t(\beta)] = 0$$

For fixed $\pi \neq 0$, it's straightforward to construct consistent, asymptotically normal GMM estimates. In weakly-identified case, we let $\pi = \pi_T = \frac{c}{\sqrt{T}}$ for $c \in \mathbb{R}^k$.

Define

$$f_T(\beta) = \frac{1}{T} \sum_{t=1}^{T} f_t(\beta)$$

therefore

$$\begin{aligned} f_T(\beta_0) &= \frac{1}{T} \sum_{t=1}^{T} [Z_t' \pi (\beta - \beta_0) + V_{1,t} - \beta_0 V_{2,t}] Z_t \\ -\frac{\partial}{\partial \beta} f_T(\beta_0) &= \frac{1}{T} \sum_{t=1}^{T} X_t Z_t = \frac{1}{T} \sum_{t=1}^{T} (Z_t' \pi + V_{2,t}) Z_t \end{aligned}$$

and let Ω be the asymptotic variance of $\sqrt{T}(f_T(\beta_0)', -\frac{\partial}{\partial \beta} f_T(\beta_0)')'$,

$$\Omega = \begin{pmatrix} \Omega_{ff} & \Omega_{f\beta} \\ \Omega_{\beta f} & \Omega_{\beta\beta} \end{pmatrix} = \lim_{T \to \infty} \mathrm{Var}\left(\sqrt{T} \begin{pmatrix} f_T(\beta_0) \\ -\frac{\partial}{\partial \beta} f_T(\beta_0) \end{pmatrix} \right)$$

Note $\Omega_{ff}, \Omega_{f\beta}, \Omega_{\beta\beta}$ are all $k \times k$ and we assume Ω_{ff} is full-rank.[2] Let $\hat{\Omega}$ consistently estimates Ω, define

$$\begin{aligned} g_T(\beta) &= \sqrt{T} \hat{\Omega}_{ff}^{-\frac{1}{2}} f_T(\beta) \\ \Delta g_T(\beta) &= -\sqrt{T} \hat{\Omega}_{ff}^{-\frac{1}{2}} \frac{\partial}{\partial \beta} f_T(\beta) \end{aligned}$$

[2]One example: if instruments are linearly dependent, then Ω_{ff} is not full-rank.

$$\hat{\gamma} = \mathrm{vec}(\hat{\Omega})$$

for $\theta = \beta$, $\Theta = \mathbb{R}$, $\gamma = \mathrm{vec}(\Omega)$, Γ is the set of values s.t. $\Omega(\gamma)$ is symmetric and positive definite. For all $(\theta, \gamma) \in \Theta \times \Gamma$, under mild conditions, we have the weak convergence and the limit problem[3]

$$\begin{pmatrix} g_T(\beta_0) \\ \Delta g_T(\beta_0) \end{pmatrix} \xrightarrow{d} \begin{pmatrix} g \\ \Delta g \end{pmatrix} \sim N\left(\begin{pmatrix} m \\ \mu \end{pmatrix}, \begin{pmatrix} I & \Sigma_{g\theta} \\ \Sigma_{\theta g} & \Sigma_{\theta\theta} \end{pmatrix} \right) \tag{37}$$

where

$$\begin{aligned} & m = \Omega_{ff}^{-\frac{1}{2}} Q_Z c(\beta - \beta_0), \qquad Q_Z = \operatorname*{plim}_{T\to\infty} \frac{1}{T} Z'Z \\ & \mu = \Omega_{ff}^{-\frac{1}{2}} Q_Z c \in \mathbb{M} = \mathbb{R}^k \\ & \Sigma_{g\theta} = \Omega_{ff}^{-\frac{1}{2}} \Omega_{f\beta} \Omega_{ff}^{-\frac{1}{2}}, \qquad \Sigma_{\theta\theta} = \Omega_{ff}^{-\frac{1}{2}} \Omega_{\beta\beta} \Omega_{ff}^{-\frac{1}{2}} \end{aligned}$$

Exercise: verify the weak convergence and the limit problem (37).

Note $m = (\beta - \beta_0)\mu$, which means $m \in \mathcal{M}(\mu) \equiv \{b\mu : b \in \mathbb{R}\}$ for any given μ. And $m = 0$ when $\beta = \beta_0$.

(ii) Minimum distance.

Parameter of interest, structural parameters $\theta \in \mathbb{R}^p$.

Reduced form parameters, $\eta \in \mathbb{R}^k$, with $\hat{\Omega}_\eta^{-1/2}(\hat{\eta} - \eta) \xrightarrow{d} \mathcal{N}(0, I)$.

The model implies

$$\eta = f(\theta)$$

so we form the estimate

$$\hat{\theta} = \arg\min_\theta (\hat{\eta} - f(\theta))' \hat{\Omega}_\eta^{-1} (\hat{\eta} - f(\theta))$$

- Under strong identification asymptotics, $\hat{\eta} - \eta = O_p(\frac{1}{\sqrt{T}})$.

[3] here the Δg itself is a $k \times 1$ vector, so no need to use $\mathrm{vec}(\cdot)$.

- If $\hat{\Omega}_\eta \xrightarrow{p} \Omega_\eta$ which is non-degenerate, then $\hat{\eta}$ is not consistent for η. The η is like the first-stage regression coefficient in weak IV setting. In this case, the nonlinearity of $f(\theta)$ will remain important even in large samples, whose weak identification corresponds to Andrews and Mikusheva (2016).

Let

$$
\begin{aligned}
&g_T(\theta) = \hat{\Omega}_\eta^{-1/2}(\hat{\eta} - f(\theta)) \\
&\Delta g_T(\theta) = -\frac{\partial}{\partial \theta'} g_T(\theta) = \hat{\Omega}_\eta^{-\frac{1}{2}} \frac{\partial}{\partial \theta'} f(\theta) \\
&\gamma = \text{vec}(\Omega_\eta), \quad \hat{\gamma} = \text{vec}(\hat{\Omega}_\eta) \\
&\Gamma = \{\gamma = \text{vec}(\Omega) : \ \Omega \text{ is symmetric p.d.}\}
\end{aligned}
$$

Note

$$
g_T(\theta_0) = \hat{\Omega}_\eta^{-1/2}(\hat{\eta} - f(\theta)) + \hat{\Omega}_\eta^{-1/2}(f(\theta) - f(\theta_0))
$$

So let $m = \lim \hat{\Omega}_\eta^{-1/2}(f(\theta) - f(\theta_0)) = \Omega_\eta^{-\frac{1}{2}}(f(\theta) - f(\theta_0))$, we have

$$
\begin{pmatrix} g_T(\theta_0) \\ \text{vec}(\Delta g_T(\theta_0)) \end{pmatrix} \xrightarrow{d} \begin{pmatrix} g \\ \text{vec}(\Delta g) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} m \\ \text{vec}(\mu) \end{pmatrix}, \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \right) \tag{38}
$$

where $m \in \mathcal{M} = \{\Omega_\eta^{-1/2}(f(\theta) - f(\theta_0)) : \theta \in \Theta\}$ and $\mu = \Omega_\eta^{-\frac{1}{2}} \frac{\partial}{\partial \theta'} f(\theta_0)$ is a constant which explains why (38) is of that variance structure. And μ clearly affects $\mathcal{M}$ for it (the first order asymptotics, slope) clearly determines the shape of $f(\cdot)$.

We then focus on the limit problem (36)

$$
H_0 : m = 0, \mu \in \mathbb{M} \qquad \text{against} \qquad H_1 : m \in \mathcal{M}(\mu) \setminus \{0\}, \mu \in \mathbb{M} \tag{39}
$$

with $(g, \Delta g, \gamma)$ being observed.

<u>Pivotal Statistics under Weak Identification</u>

- $S = g'g \sim \chi^2_k(m'm)$.

- Define D as the $k \times p$ matrix s.t.

$$\text{vec}(D) = \text{vec}(\Delta g) - \Sigma_{\theta g} g$$

and note

$$\begin{aligned}
&\text{Var}[\text{vec}(D)] = \Sigma_{\theta\theta} - \Sigma_{\theta g}\Sigma_{g\theta} - \Sigma_{\theta g}\Sigma_{g\theta} + \Sigma_{\theta g}\Sigma_{g\theta} = \Sigma_{\theta\theta} - \Sigma_{\theta g}\Sigma_{g\theta} \equiv \Sigma_D \\
&\text{Cov}[g, \text{vec}(D)] = \Sigma_{g\theta} - \Sigma_{g\theta} = 0
\end{aligned}$$

so

$$\begin{pmatrix} g \\ \text{vec}(D) \end{pmatrix} \sim N\left(\begin{pmatrix} m \\ \text{vec}(\mu_D) \end{pmatrix}, \begin{pmatrix} I & 0 \\ 0 & \Sigma_D \end{pmatrix} \right)$$

where $\text{vec}(\mu_D) = \text{vec}(\mu) - \Sigma_{\theta g} m$, let $\mu_D \in \mathbb{M}_D$, and we write

$$m \in \mathcal{M}_D(\mu_D) \equiv \{m : m \in \mathcal{M}(\mu) \text{ for } \text{vec}(\mu) = \text{vec}(\mu_D) + \Sigma_{\theta g} m\}$$

here $\mathcal{M}_D$ plays a role similar to $\mathcal{M}$, defining the set of values m consistent with a given mean μ_D for D.

$$K = g'D(D'D)^{-1}D'g = gP_D g \tag{40}$$

and

$$J = S - K = g'(I - D(D'D)^{-1}D)g = g'(I - P_D)g \tag{41}$$

<u>We focus on class of tests written as function of S, K and D.</u>

- Conditional on $D = d$ for d full column-rank, note $D \in \mathbb{R}^{k \times p}$, this implicitly impose $k \geq p$, actually meaningful with strictly $k > p$, which implies the over-identified models: more moments than number of parameters to be estimated.

- Important insights:

- Go back to the minimum distance example (continued).

  Δg is nonrandom, so $\Sigma_{\theta g} = 0$, $D = \Delta g = \mu = \mu_D$. Assume $\Omega_\eta = I$ to simplify the exposition, then

$$(\tau_J, \tau_K)$$

- Go back to the weak instrument example (continued).

  In this example, Δg is random and can be correlated with g, $D = \Delta g - \Sigma_{\theta g} g \in \mathbb{R}^{k\times 1}$ and $\mu_D = \mu - \Sigma_{\theta g} m$ where $m = \mu(\beta - \beta_0)$ and $\mu = \Omega_{ff}^{-\frac{1}{2}} Q_Z c$, so

$$\mu_D = \mu - \Sigma_{\theta g}\mu(\beta - \beta_0) = (I - \Sigma_{\theta g}(\beta - \beta_0))\mu$$

  if μ is proportional to a eigenvector of $\Sigma_{\theta g}$ corresponding to a nonzero eigenvalue λ,

$$\lambda\mu = \Sigma_{\theta g}\mu$$

  then

$$\mu_D = \mu - \lambda\mu(\beta - \beta_0)$$

  If $\beta - \beta_0 = \lambda^{-1}$, then $\mu_D = 0$. Note $K = g' P_D g = \frac{(D'g)^2}{D'D}$, under the null $K \sim \chi^2(0)$, , the well-known non-monotonicity of the power function for tests based on K.

5.7 Summary

Tests: AR, LM, CLR, CLC, MM1, MM2, CIL. Conditions: LU, SU.

6 Test for weak instrument

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